

GDSS-PSInno AgriTech Innovation Challenge

Challenge Title: Farmer-to-Buyer Digital Marketplace Platform

This challenge is organized by Ghana Data Science Summit (GDSS) and sponsored by GIZ under the Private Sector and Innovation Promotion in Ghana (PSInno) programme.

Prizes

- 1st Prize: GHS 10,000
- 2nd Prize: GHS 7,000
- 3rd Prize: GHS 5,000

Problem

Agriculture remains one of the most important sectors of Ghana's economy, contributing approximately 20% of the country's GDP and employing more than 30% of the national workforce. The sector is dominated by smallholder and small-to-medium scale farmers, with over 80% of farms in Ghana operating on less than five acres of land.

Despite its importance, small-to-medium scale farmers growing perishable vegetables, particularly tomatoes, peppers, garden eggs, okra, and leafy greens face challenges in accessing reliable markets and fair pricing. The time-sensitive nature of perishable produce makes inefficiencies in market access especially costly

In addition, even when farmers successfully connect with buyers, transporting produce efficiently and affordably remains a major challenge. Poor logistics coordination, lack of reliable transport options, and fragmented delivery systems often lead to delays, spoilage, increased costs, and lost income.

Ghana is estimated to lose between 20% and 50% of fruits and vegetables after harvest due to poor storage systems, transportation delays, and weak distribution networks. These losses are estimated to cost the country between \$1.9 billion and \$3 billion annually.

At the same time, buyers such as retailers, restaurants, processors, exporters, and households struggle to identify trusted suppliers, compare prices, locate nearby produce, and coordinate timely delivery of goods.

For the purposes of this challenge, participants are encouraged to focus on a specific agricultural supply regions rather than attempting to solve the problem for the entire country.

Proposed Solution

This project aims to develop a digital platform that connects smallholder vegetable farmers directly to buyers through a mobile or web application, while also enabling efficient coordination of transport and delivery.

The proposed digital platform may include (but not limited to) the following capabilities/features:

- Mobile application marketplace platform
- Web-based marketplace platforms
- USSD/SMS integrations for low-connectivity areas
- Mapping and geolocation systems
- Digital payment integrations
- Notification and communication systems
- Logistics and delivery coordination systems
- Transport provider matching and scheduling tools
- Data-driven recommendation or matching systems

Core Features

The platform may include some or all of the following features:

Farmer Features

- Farmer registration and profile creation
- Produce listing with images, quantity, location, and pricing
- Inventory updates and availability tracking
- Communication with buyers
- Ability to request transport or delivery services

Buyer Features

- Buyer registration and profile creation
- Search and filtering of products by category, price, location, or quantity
- Direct ordering or negotiation with farmers
- Order history and transaction tracking
- Ability to coordinate or track delivery status

Logistics Features

- Transport provider registration and profile creation
- Transport request and booking system
- Matching of farmers and buyers with nearby transport providers
- Delivery scheduling and route coordination

- Estimated transport cost calculation
- Delivery tracking and status updates
- Notifications for pickup and drop-off events

Other generic platform features

- Product listing and marketplace interface
- Geolocation or mapping of farmers, buyers, and transport providers
- In-app messaging or contact system
- Notifications for new listings, orders, and deliveries
- Rating and review system for farmers, buyers, and transport providers
- Mobile money or payment integration
- Dashboard for monitoring transactions, logistics, and activity

Key factors in assessing the winning solution

- Accessibility and usability for farmers, buyers, and transport providers
- Scalability and sustainability of the solution
- User-friendly interface and workflow
- Efficient product discovery and transaction flow
- Effective logistics coordination and delivery tracking
- Innovation and practical relevance
- Potential for real-world adoption and impact
- Relevance to the target vegetable supply chain
- Offline or low-bandwidth considerations (bonus)
- Integration of smart recommendation or matching systems (bonus)

Key Deliverables

Each team is expected to deliver:

- A working prototype or MVP
- A frontend interface (mobile or web)
- A backend/database system
- A live demo of the platform workflow that includes the following
 - Farmer registration and produce upload
 - Buyer product search and interaction
 - End-to-end marketplace operations workflow
 - Additional innovative features implemented by the team
 - Transport request and delivery/logistics coordination flow (bonus)

Datasets and Supporting Materials

Teams may use publicly available datasets and APIs where applicable. This may include agricultural produce data, pricing datasets, geolocation data, and transport-related datasets where available.

Stakeholder Considerations

Participants are encouraged to consider the needs of stakeholders such as farmers, buyers, and transport providers in designing their solutions. Light user research or informal consultations are encouraged but not required.

Timelines

Activity	Start Date	End Date
Registration deadline		26 June 2026
Challenge period	29 June 2026	10 July 2026
First round of judging and shortlisting	13 July 2026	17 July 2026
Final presentation and judging		24 July 2026
Awards and prize presentation		24 July 2026

Contact us via programs@datasciencenet.org if you have any questions.